C AIE Scores

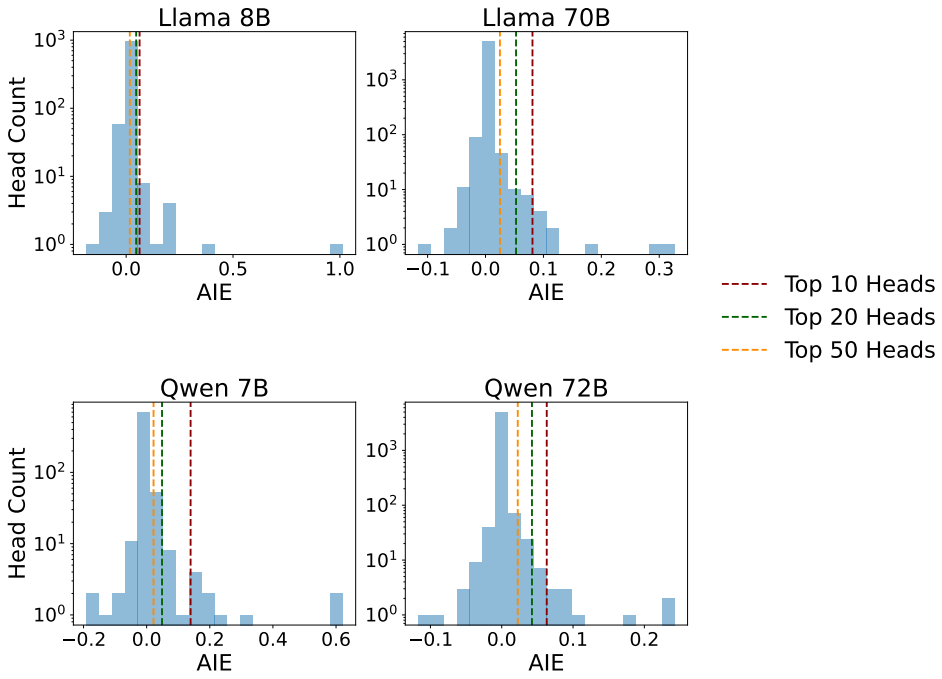

Figure 12: Histogram of AIE scores for Llama 3.1 8B, 70B, Qwen 2.5 7B, and Qwen 2.5 72B. Note, the y-axis is on a log-scale. **Takeaway**: AIE scores are highly sparse.

D Data Generation Process

Concept sourcing: For most concepts (antonym, synonym, translation, present–past, singular–plural), we sourced word pairs from the datasets used by Todd et al. (2024). For categorical and causal concepts, we generated word pairs using OpenAI's GPT-4o model (OpenAI, 2024).

Translation generation: French and Spanish translations were created using DeepL's translation service (DeepL SE, 2025) to ensure high-quality, contextually appropriate translations.

Generated concepts (categorical and causal): We prompted GPT-4o to generate exemplar:category pairs (e.g., "apple:fruit", "blue:colour") and cause:effect pairs (e.g., "stumble:fall", "storm:flood"). The model was given examples of the desired format and asked to produce 100 pairs per batch. We generated pairs in batches of 100 until reaching approximately 1000 examples per concept, with retry mechanisms to ensure sufficient coverage. The final datasets were saved as JSON files containing input-output pairs.

Quality filtering: Generated pairs underwent several filtering steps: (1) removal of duplicates based on input words, (2) exclusion of pairs containing underscores or numbers, (3) restriction to single words or two-word phrases (maximum one space per input/output), and (4) conversion to lowercase for consistency.

Multiple choice format: For multiple choice prompts, we generated four options per question by randomly sampling three additional outputs from the same concept dataset, ensuring all four options were unique. The correct answer was randomly positioned among the four options.

E Significance Test for RSA–AIE Head Overlap

We assess whether the observed overlap between the top-K heads selected by Concept-RSA and by AIE is larger than expected by chance under a simple null model. Let N denote the total number of attention heads in the model (layers $\times$ heads per layer). For a fixed K, each method selects a size-K subset of heads. Under the null hypothesis that these two subsets are independent, uniformly random size-K subsets of $\{1, \ldots, N\}$, the overlap size

$$X = |S_{\text{RSA},K} \cap S_{\text{AIE},K}|$$

follows a hypergeometric distribution $X \sim \text{Hypergeom}(N, K, K)$.

For an observed intersection x, we report the one-sided tail probability

$$p_{\geq x} = \Pr\left[X \geq x\right] = \sum_{t=x}^{K} \frac{\binom{K}{t}\binom{N-K}{K-t}}{\binom{N}{K}} .$$

Entries with $p_{\geq x} < 0.05$ are typeset in bold in Table 1.

F Steering Hyperparameters

To optimize the intervention performance, we conduct a hyperparameter search for two parameters:

- α: the steering weight that controls the strength of the intervention
- K: the number of attention heads to extract for concept vector computation

We evaluate the following parameter ranges:

- $K \in \{1, 3, 5, 10, 20, 50\}$ for the number of heads
- $\alpha \in \{1, 3, 5, 10, 15\}$ for the steering weight

The hyperparameter optimization is performed separately for each model using antonym prompts. We select the parameter combination that maximizes the average steering effect across all input formats. This ensures that our chosen hyperparameters generalize well across different prompt structures. We report the best hyperparameters for each model in Table 3.

Model	**Best** K	**Best** α
Llama 3.1 8B	1	10
Llama 3.1 70B	5	10
Qwen 2.5 7B	3	10
Qwen 2.5 72B	5	15

Table 3: Optimal hyperparameters for steering interventions across different models. K represents the number of attention heads used for $\mathcal{FV}$/$\mathcal{CV}$extraction, while α controls the intervention strength.

G INPUT FORMAT MIXING IN FUNCTION VECTORS

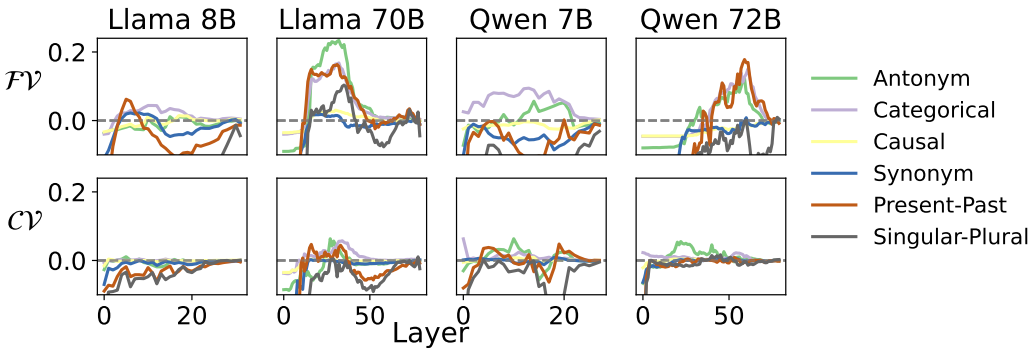

Figure 13: ΔP for French translations of all the concepts. $\mathcal{FV}$s and $\mathcal{CV}$s are extracted from open-ended French prompts.

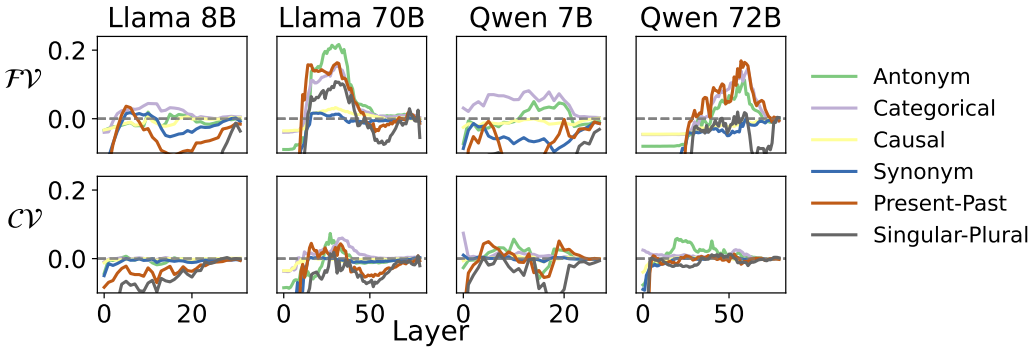

Figure 14: ΔP for French translations of all the concepts. $\mathcal{FV}$s and $\mathcal{CV}$s are extracted from open-ended Spanish prompts.

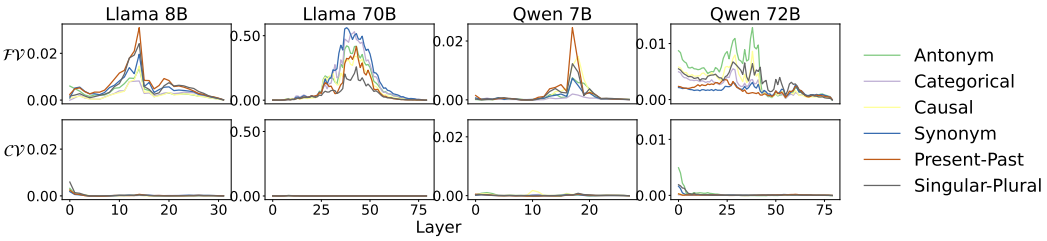

Figure 15: ΔP for the opening bracket token _(. $\mathcal{FV}$s and $\mathcal{CV}$s are extracted from multiple-choice prompts.